

WEEK 3 ASYNCHRONOUS ACTIVITY

Profiling by counting · data types · applied steps as documentation

Why this is asynchronous	Saturday's face-to-face session is suspended for weather.
What replaces it	A one-hour online discussion, then this activity on your own.
How long it should take	About two and a half hours. Not one sitting — take breaks.
Due	Wednesday, 2 September 2026, 11:59 PM
Where to submit	Your b2-cleaned-dataset folder in BDAT04-2026_T1

The one rule that makes this work. Every exercise asks you to write down what you EXPECT before you measure it. Do not skip that. A guess you committed to and got wrong teaches you something; a number you read off a screen teaches you nothing. I will be reading your predictions, and being wrong costs you no marks whatsoever.

Before you start

1. Open app.powerbi.com and sign in with your LPU account.
2. Open your semantic model, then click **Transform data** to reach the Power Query editor.
3. Confirm the Queries pane on the left says **Queries [7]** and lists all seven tables. If it does not, stop and message me.

If Power BI will not load. Some accounts are affected by a Microsoft rollout problem this week. It is not your fault and there is nothing you can click to fix it. Use the Excel version instead — open the case workbook, click inside any table, and use Data → From Table/Range. The editor is the same, the exercises are identical, and your answers will be the same. Tell me you did it that way when you submit.

Part A — Counting to find what your eyes cannot

In the discussion I showed you this on **BranchID**. Now do it yourself, on four columns I did not show you.

For each one: **select the column → Transform tab → Group By → Operation: Count Rows → OK**. Then remove the step and move on.

A1 — Predict first

Before you touch Power Query, fill in the middle column from what you already know about the business. Do not open the file. Guess.

Table	Column	You predict	You measure	Verdict
tblInventory	MovementType	I predict that there would only be 1 category.	There are <u>5 measured</u> categories in total.	The data shows that there is 1 inconsistent metric “delivery” should be “Delivery”. It shows a reading of 5 categories in total, while there should only be 4 due to an error in inputting the MovementType.

				Visualization: <table><tr><th></th><th>MovementType</th><th>Count</th></tr><tr><td>1</td><td>Delivery</td><td>1451</td></tr><tr><td>2</td><td>Wastage</td><td>23</td></tr><tr><td>3</td><td>Adjustment</td><td>23</td></tr><tr><td>4</td><td>delivery</td><td>25</td></tr><tr><td>5</td><td>Transfer Out</td><td>19</td></tr></table>		MovementType	Count	1	Delivery	1451	2	Wastage	23	3	Adjustment	23	4	delivery	25	5	Transfer Out	19
	MovementType	Count																				
1	Delivery	1451																				
2	Wastage	23																				
3	Adjustment	23																				
4	delivery	25																				
5	Transfer Out	19																				
tblCustomers	Tier	I predict that there would be 3 categories.	There are <u>5 measured</u> categories in total.	The data shows that there are 2 inconsistent metric, “REG” should be listed as “Regular” and “silver” should be listed as “Silver”. It shows a reading of 5 categories in total, while there should only be <u>3 categories</u> due to an error in inputting the Tier. Visualization: <table><tr><th></th><th>Tier</th><th>Count</th></tr><tr><td>1</td><td>Regular</td><td>101</td></tr><tr><td>2</td><td>Silver</td><td>54</td></tr><tr><td>3</td><td>Gold</td><td>13</td></tr><tr><td>4</td><td>REG</td><td>16</td></tr><tr><td>5</td><td>silver</td><td>2</td></tr></table>		Tier	Count	1	Regular	101	2	Silver	54	3	Gold	13	4	REG	16	5	silver	2
	Tier	Count																				
1	Regular	101																				
2	Silver	54																				
3	Gold	13																				
4	REG	16																				
5	silver	2																				
tblSales	PaymentMethod	I predict that there would be 4 categories. (e.g. cash, fintech, credit card)	There are <u>3 measured</u> categories in total.	The data is clean and consistent. No revisions are required.																		
tblProducts	Category	I predict that there would be 5 categories.	There are <u>5 measured</u> categories in total.	The data is clean and consistent. No revisions are required.																		

Now measure. Fill the fourth column. In the fifth write **clean** or list the offending values exactly as they appear.

Not every column is dirty. Two of those four are exactly as they should be. Proving a column is clean is a real finding, and you will need to say so in a report one day. Anyone who reports four problems has stopped counting and started assuming.

A2 — Counting what is missing

Click the ▼ arrow on `tblSales[CustomerID]`, untick (Select All), tick only (`null`), OK.

Prediction — what fraction of sales have no customer recorded? 40%

Measured — how many rows? 4775 out of 10,589. That is 45%

Now the question that matters, and the one I want a written answer to:

Is this a data quality defect, or a fact about how Kape Tayo does business? Answer in two or three sentences. Then say what you would do about it — and if your answer involves deleting rows, say how much of the company's revenue you just deleted.

Answer:

Based on the data provided, this reading shows that a total of **45% customers of Kape Tayo** represent walk-in customers who do not have or applied for a customer account. This reading does not necessarily suggest a data-quality defect and simply deleting the data would render their sales data irrelevant which would cause a bigger data issue.

My suggestion would be to keep the data and I would document that a “null” CustomerID represents walk-in transactions.

A3 — Counting duplicates

1. Select `tblSales[SalesID]` → Home → Keep rows → Keep duplicates.
2. Record the number of rows shown: 46
3. Remove that step. Now Home → Remove rows → Remove duplicates. Record the new row count: 10,566
4. Subtract from 10,589. Surplus rows: 23

Two numbers, two different questions. In one sentence each: what does the first number answer, and what does the second answer? Which one would you put in a report recommending a fix?

1. The first number tells the number of rows involved in duplicate sales ID in the data.
2. The second number shows how many duplicate sales ID should be removed to render the data relevant.
3. I would report the 23 surplus rows because this represents the actual number of duplicate records that would need to be removed.

Part B — Data types, and the one that bites

Look at the small icon left of each column name. **ABC** is text. **123** is a whole number. A column typed as text will not sort, will not total, and will not join to your calendar. Most modelling failures are type failures in disguise.

B1 — The date that is not a date

1. In tblSales, click the icon on **Date** and choose **Date**.
2. Watch Applied Steps. A step appears. Watch the formula bar.
3. Filter the column for **(null)** and count. Rows: **12 rows**.

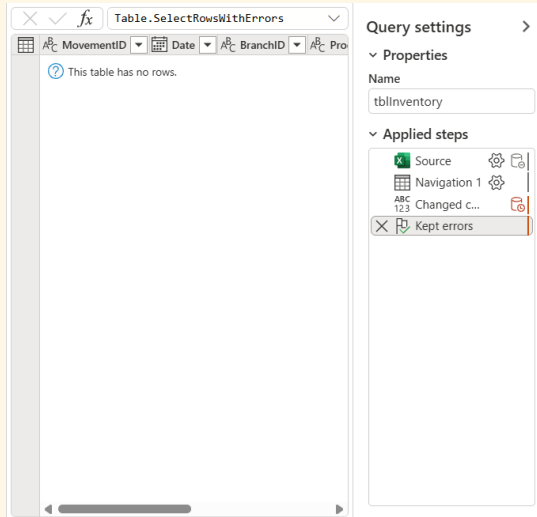
A type change that produces no error message is not proof of success. It can quietly produce blanks instead. Counting after a type change is a habit worth keeping.

B2 — The one that breaks

Go to **tblInventory** and change **QuantityMoved** to Whole Number.

Rows will turn into Error. Do NOT click Remove Errors. Instead: Home → Keep rows → Keep Errors. Look at what those values actually are.

When I initiated these commands, it showed no data. See Figure 1.



(Figure 1)

Instead, I initiated a **group by** command, and sorted these values by **counts of 1**. Which showed the irregularities in the data. See Figure 1.1.

	QuantityMoved	Count
1	4195	1
2	1582	1
3	4404	1
4	4663	1
5	2022	1
6	-15	1
7	2426	1
8	3382	1
9	4301	1
10	1995	1
11	2982	1
12	4261	1
13	1796	1
14	2990	1
15	2423	1

(Figure 1.1)

Errors found: **14 rows** What the offending values look like: **Numbers containing thousands.**

Write down why a number refused to become a number. Then work out the correct order of operations to fix it — one of these two sequences works and the other fails: The values containing commas as thousand separators caused the error. Instead of using the said operating procedure, I compromised based on my knowledge and did the operation in this order:

groupby > sort drop down button > unselect all > select “1” > OK

Sequence 1	Sequence 2
Change type → Replace Values	Replace Values → Change type

Which works, and why? One sentence:

Neither one of these sequences worked for my Power Query.

Then the part that should worry you: if you had clicked Remove Errors, the query would still run, no warning would appear, and no step name would say what happened. Work out which rows you would have lost and what they have in common. That answer goes in your reflection.

B3 — Arithmetic that does not agree

1. In tblSales: **Add Column → Custom Column**, name it **Check**:

$$= [\text{Quantity}] * [\text{UnitPrice}] - [\text{Amount}]$$
2. Filter it to anything not equal to 0. Count the rows: **23 rows**
3. Decide which of the three columns is wrong — and whether you should change anything at all.

There is no single right answer here. There is a right *behaviour*: report it, do not silently overwrite it. Say what you would tell the branch manager.

From this calculation alone, I will not be able to determine which of these rows are incorrect. Overwriting these values without any additional evidence is not ideal. I would tell the branch manager that 23 transactions have Amount values that do not reconcile with Quantity × UnitPrice and ask that they be checked against the original transaction records before making any corrections.

Part C — Applied steps as documentation

Your Applied Steps list probably reads: Source, Navigation, Changed Type, Filtered Rows, Changed Type1. That is a machine talking to itself, not documentation.

Right-click each step → **Rename**. Say what it does and, where it is not obvious, how many rows it touched.

Instead of	Write something like
Changed Type	Set Date to date type
Filtered Rows	Removed 12 rows with no sale date
Removed Duplicates	Removed 23 duplicate SaleIDs
Replaced Value	Standardised BranchID casing and spacing

Then right-click any step that involved a judgement → **Properties** → write one sentence explaining *why* you were entitled to do it. Every step that deleted rows needs one.

This is the part that is graded most heavily. Rubric D gives 20% to documentation and readability. More usefully: you will be the one reading this in Week 11, at night, when something has broken and you cannot remember what you did in August.

Part D — Back it up

TMDL View is a blank scratchpad. It does not show your model until you put it there, and it keeps nothing when you close it. That is why you save the text yourself.

1. Bottom of the model window → **TMDL View**.
2. Open the **Data pane** on the right and DRAG a table into the editor. Its script appears. Repeat for each table.
3. Click in the editor, select all, copy.
4. Paste into a plain-text editor. **Windows: Notepad. Mac: TextEdit, but press T first to make it plain text.**
5. Save as `model_<date>.tmdl` and upload to your **backups** folder.

Never click Apply. Apply pushes whatever is in the editor INTO your model. It is for making changes, not saving them. For backup you only ever read and copy.

Part E — Written reflection

This replaces the discussion we would have had in the room. Half a page is plenty. Write it in `reflection.md` in the same folder.

- How many rows did `tblSales` start with, and how many does it have now? For every row you removed, can you name the reason?
- Which of your decisions was a judgement call rather than a correction?
- Which finding would you put in front of a branch manager, and which would you fix quietly? Why the difference?
- **What did you find today that you missed completely in B1?**

That last one is the point of the whole week. Nothing in the data changed between B1 and today. Only your method did. Say what changed.

What to submit

Item	Detail
Answers	This sheet with every blank filled — scan, photograph, or retype. Predictions included.
<code>reflection.md</code>	Part E, in your own words
model TMDL export	<code>model_<date>.tmdl</code> in your backups folder
Renamed steps	Visible in your saved model — I will open it
AI Use Log	Updated, per the course policy. Tier and purpose.

Upload to `b2-cleaned-dataset` in `BDAT04-2026_T1`. Commit message: `W3 async activity — <your surname>`

If you get stuck

Fifteen minutes, then message me. Not an hour. Not Sunday night. There is no prize for suffering alone, and a question at 3 PM on Saturday costs me two minutes. Tell me the exercise number, what you clicked, and what happened. A screenshot is better than a description.

I will be checking messages through Saturday afternoon and Sunday morning.

Self-check — do not read until you have finished

Finish everything above first. If you read these numbers before measuring, you will not learn the thing this activity exists to teach, and I will be able to tell from your predictions.

If a number of yours differs, do not just correct it. Find out why — usually a filter still applied, or the wrong table selected. That hunt is worth more than the number.

Part A

Column	Answer	What is there
MovementType	5	Should be 4. Delivery 1,451 · delivery 25 · Wastage 23 · Adjustment 23 · Transfer Out 19
Tier	5	Should be 3. Regular 101 · Silver 54 · Gold 13 · REG 16 · silver 2
PaymentMethod	3	Clean. Cash 5,884 · GCash 3,382 · Card 1,323
Category	5	Clean
Blank CustomerID	4,775	45.1% of sales. NOT a defect — walk-in customers
Duplicate SaleIDs	46 / 23	46 rows involved, 23 surplus, 10,566 distinct

On Tier: REG is not a casing error. It is a different vocabulary, and Lowercase or Trim will not touch it. Somebody has to decide REG means Regular — and that decision belongs in a step description, not in a silent Replace Values.

Part B

Check	Answer	Note
Blank dates after type change	12	Nulls, not errors. No warning appears.
QuantityMoved errors	14	Thousand separators. These are the largest deliveries in the table.
Correct order	Sequence 2	Replace Values first, then change type. The reverse fails.
Amount mismatches	23	Ambiguous by design. Report it; do not overwrite it.

The QuantityMoved trap: Remove Errors would have silently deleted the fourteen largest deliveries in the dataset. Inventory totals fall, the query still runs, and no step name says why. If you did click it — good. You now know something the others only read about.

Questions on any of this? Message me. I would rather answer on Sunday than mark a misunderstanding on Monday.

— Sir JB